Open Chameleon app

Sign in
Sign up
Forgot password

User lands on dashboard

Open theme settings

Select Dark /
Light / Auto

Interface updates accordingly

Save Preferences for next time

Ambient light changes

System detects light conditions

Light /
Intermediate /
Dark theme
applied

Interface adapts
in real time

Adaptive Dashboard

Normal task dashboard

Minimal distractions + Active task/timer

No tasks

Highlight overdue tasks

Streaks,
Completion %,
Achievements

Open accessibility options from profile/settings

1.Small
2.Medium
3.Large
4.Extra Large

1.Better visibility
2.Adaptive colors

1.Task narration
2.Alerts/ reminders
3.Read screen content

→ Show cached tasks/ progress
→ Allow limited functionality
→ Retry when connection returns.

→ Explain the problem
→ Retry
→ If needed, show cached/last updated data.

If the device already has accessibility settings enabled:

Detect system settings → automatically adapt the UI.

→ Session timeout
→ Invalid data
→ Unexpected error

The dashboard automatically changes its layout, text size, spacing, etc.

Limited hand movement / precise checking

Difficulty reading small text

Dark room, bright light

Distraction, need for simplified UI

Preference, or inability to use touch

Motor difficulty detected

Increase hit target size (bottons, cards, icons

Larger interactive elements applied

User interacts easily with controls

Low vision or text size needed

Increase text size and adjust typography

Updated layout (prevent overlap)

User reads and interacts comfortably

Low light / high light environment or contrast need detected

Switch to high contrast / adaptive colours

Interface updates with better visibility

User continues with task with improved visibility

Voice assistance need detected

Enables voice prompts and audio feedback

Show listening state and feedback

User interacts using voice commands

High cognitive lead detected

Reduce distractions & simplify UI

Essential elements highlighted

User completes task with minimal effort



High ambient light detected

↓

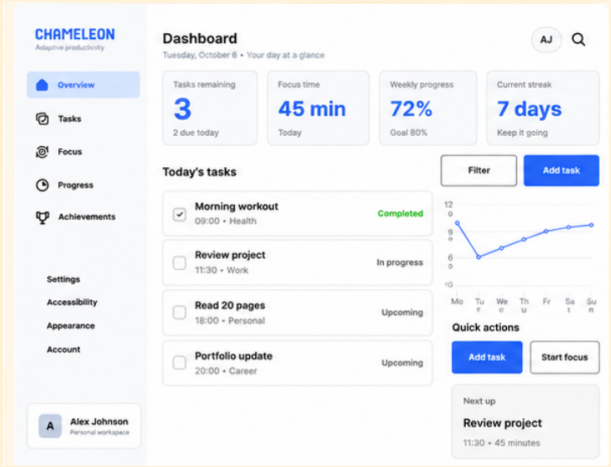
Apply light theme (bright background, dark text)

↓

Maintain accessibility settings

↓

Interface updated for light mode



Bright background, dark text
Clear and high visibility

Moderate / changing light detected

↓

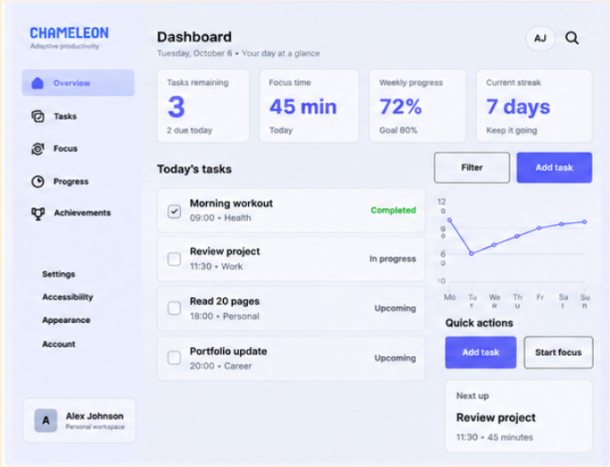
Apply intermediate theme (balanced contrast)

↓

Adjust colour and UI elements for readability

↓

Interface updated for intermediate mode



Balanced contrast
Adapts smoothly too changing lights

Low ambient light detected

↓

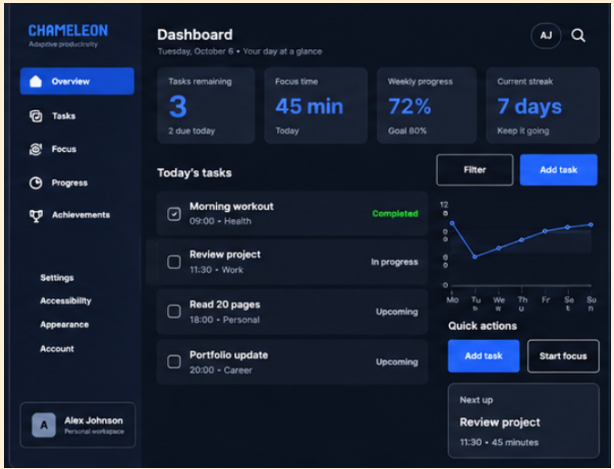
Apply dark theme (dark background, light text)

↓

Maintain accessibility settings

↓

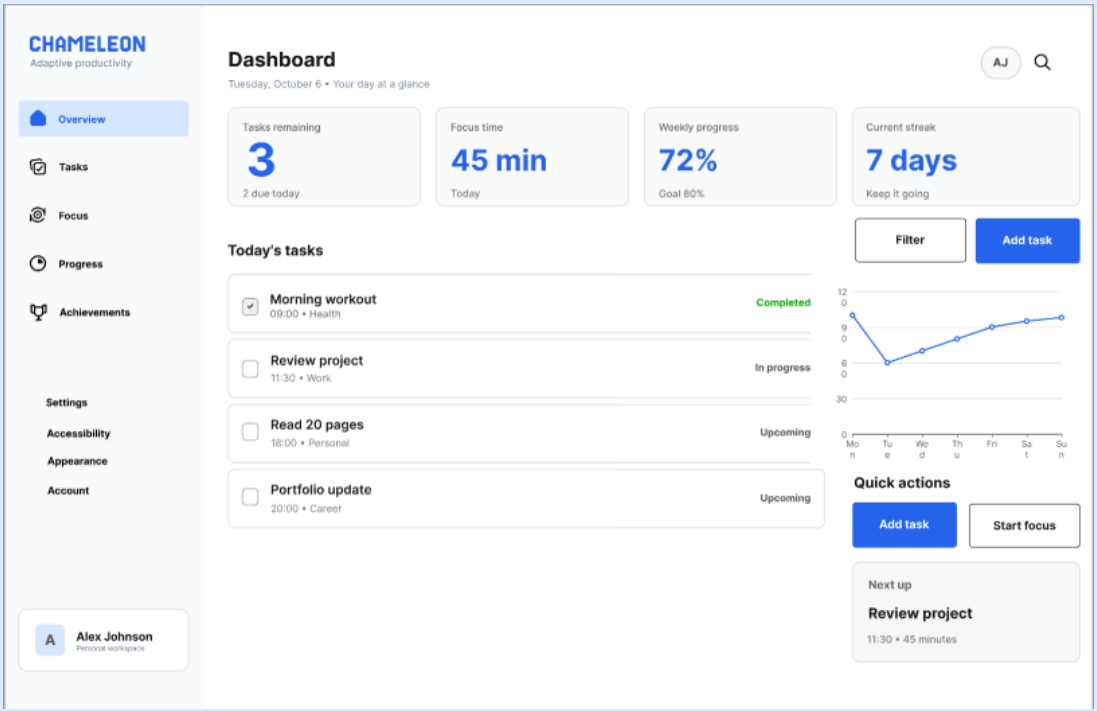
Interface updated for dark mode



Dark background, light text
reduced glare, better visibility

User on Dashboard

User opens the Chameleon dashboard with their current theme and accessibility settings.



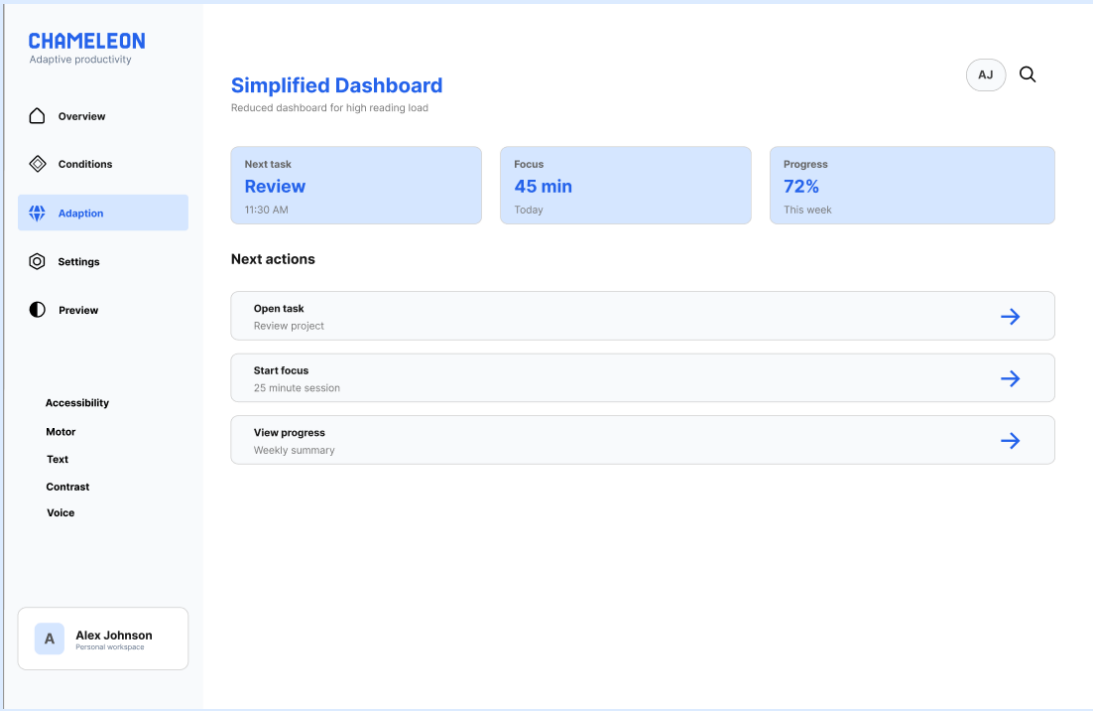
Need for a Simpler Interface

User finds the interface visually overwhelming or distracting and needs a simpler view.

- Too much information
- Distracting elements
- Difficult to focus
- Need a simpler view

Reduce Visual Clutter

Non-essential elements are reduced and information is organized into a clearer layout.



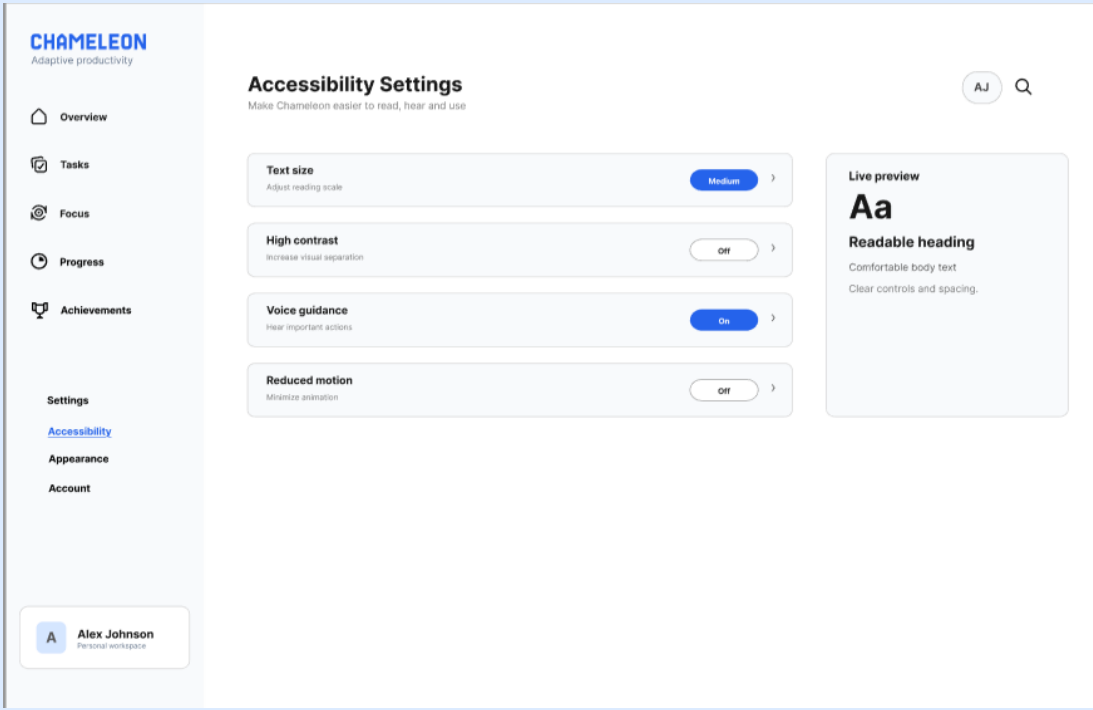
Important Information Highlighted

Important information and primary actions are made easier to identify.

- Highlight important tasks.
- Make primary actions more prominent.
- Improve information hierarchy.
- Keep secondary information accessible.

Reduced Motion

User can enable reduced motion to minimize non-essential animations and motion effects.



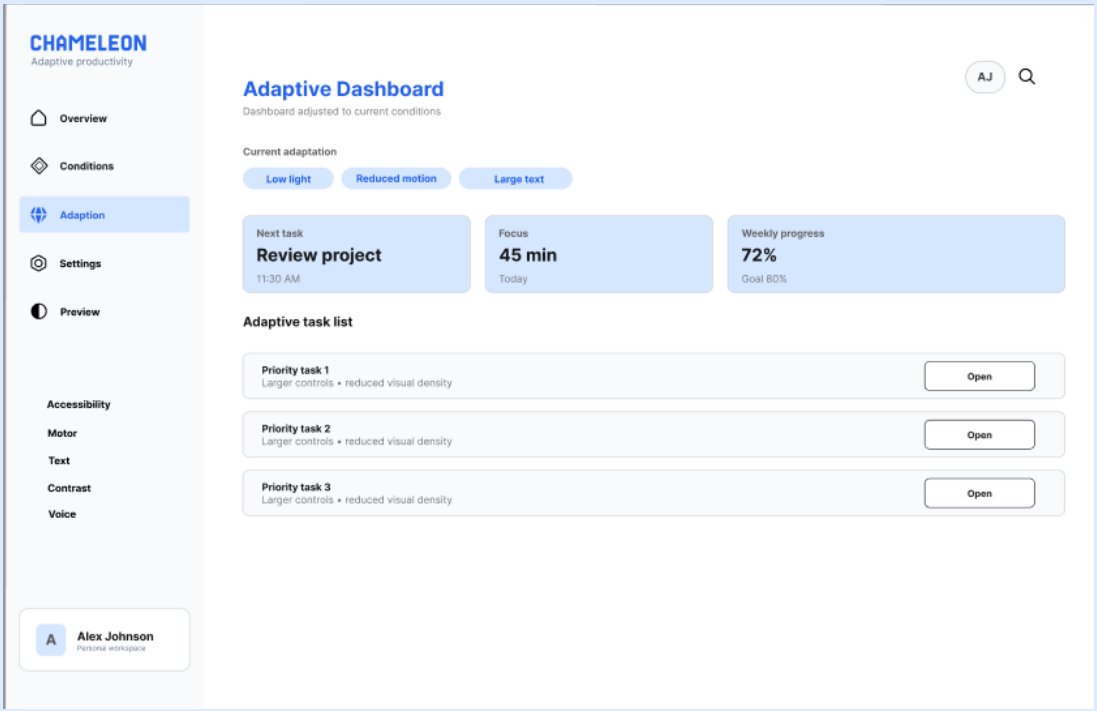
Accessibility Check

Check that the adapted interface remains accessible and easy to use.

1. Text Readability – Text size and contrast remain readable.
2. Interactive Target Sizes – Buttons and interactive elements are easy to select.
3. Layout Clarity – Simplified layout maintains a clear information hierarchy.
4. Motion Preferences – Reduced motion settings are respected.
5. Colour Contrast – Text and controls remain distinguishable in both light and dark themes.

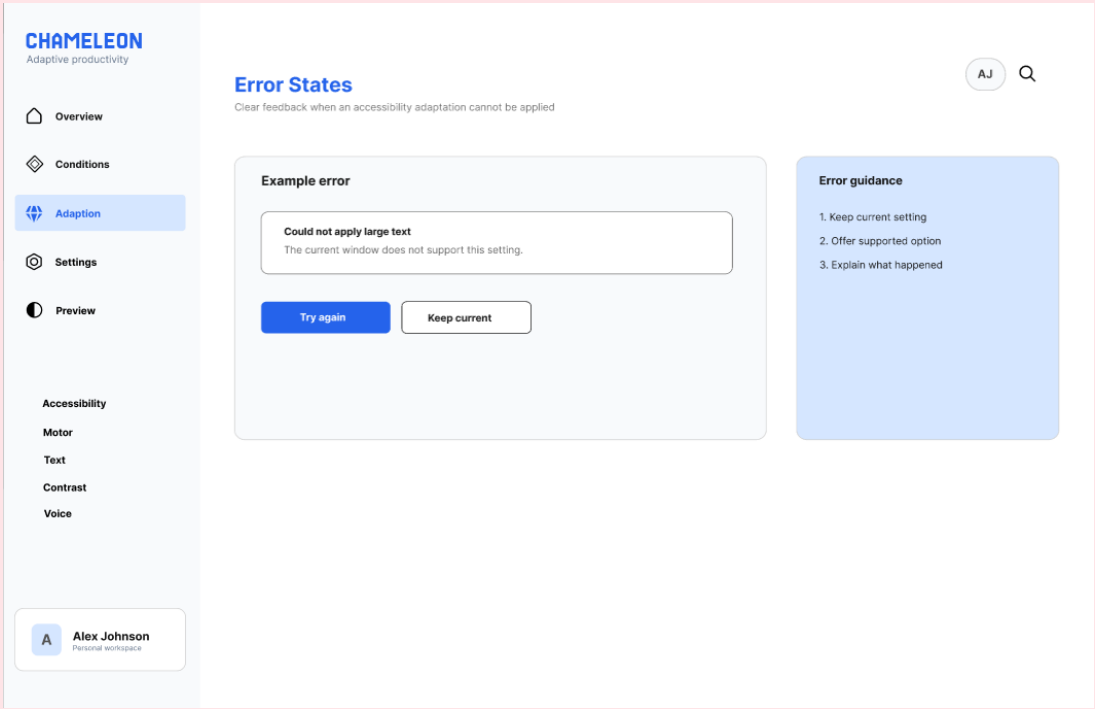
Review Adapted Dashboard

User reviews the simplified interface and continues their task.



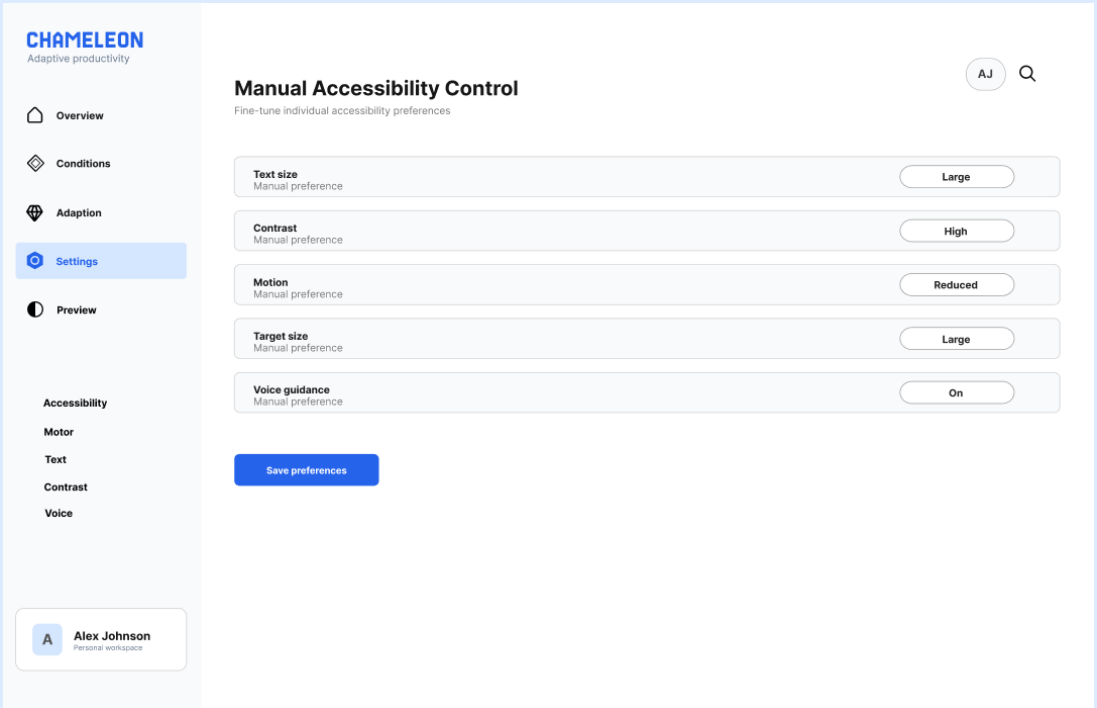
Unable to Apply Adaptation

The system could not apply the requested accessibility adaptation. The user can adjust the settings manually.



Customize Accessibility Preferences

User can manually adjust available accessibility preferences.



User Continues Task

The user continues their task using an interface that is simpler and more accessible.

- Reduced distractions.
- Accessibility settings preserved.
- User can adjust preferences manually.
- Consistent experience across Light and Dark themes.